Name ______________________________ Section ________

MTH:2310 DISCRETE MATH
QUIZ 2
DUE: TUESDAY FEB. 10

Complete all of the following problems being as detailed as you can be. You may work in groups, but you must submit your own solutions using your own words. You must submit a physical copy of your solutions to me unless told otherwise. If there is a specific number of problems which seem unsolvable let me know first, I may have made a typo. Even if you cannot completely solve a problem, a good description of key terms and concepts relevant to the problem is sufficient for some partial credit.

Handwriting: (4 points)
Your ability to express your solutions is just as important as mathematical accuracy.
You will be penalized up to 2 points for arithmetic errors.
You will be penalized up to 2 points for disorganized or hard to read solutions.

Problem 1. *(8 points) De Morgan's Law*
Identify and define all atomic propositions in the following statements. Use De Morgan's law to state their *negation* as a compound proposition, and then as a sentence.

(a) *(4 points) "Zach owns a 2009 Honda Civic and he drives it every day."*

p = Zach owns a 2009 Honda Civic.

q = Zach drives his car every day.

$\sim(p \wedge q) \equiv \sim p \vee \sim q$

Zach does not own a 2009 Honda Civic or he does not drive it everyday.

(b) *(4 points) "Zach drinks coffee or an Alani energy drink in the morning."*

p = "Zach drinks coffee in the morning."

q = "Zach drinks an Alani energy drink in the morning."

$\sim(p \vee q) \equiv \sim p \wedge \sim q$

Zach does not drink coffee and he does not drink an Alani energy drink in the morning.

Problem 2. *(8 points) Logical Equivalences*
Use a truth table to determine whether the following compound propositions are logically equivalent.

(a) *(4 points)* $\Big(p \vee (p \wedge q)\Big) \stackrel{?}{\equiv} p$

p	q	$p \wedge q$	$p \vee (p \wedge q)$
T	T	T	T
F	T	F	F
T	F	F	T
F	F	F	F

They are equivalent

(b) *(4 points)* $\Big(p \Rightarrow q\Big) \stackrel{?}{\equiv} \Big(\sim p \vee q\Big)$

p	q	$\sim p$	$p \Rightarrow q$	$\sim p \vee q$
T	T	F	T	T
F	T	T	T	T
T	F	F	F	F
F	F	T	T	T

They are equivalent

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Problem 3. *(10 points) Logical Equivalences*
Show that the following compound propositions are equivalent using a series of logical equivalences.
If you use an equivalence from Table 6, 7, or 8 (pg. 29), reference the specific table. Otherwise, you must fully justify the logical equivalence.

(a) *(5 points)* $\Big((p \Rightarrow \sim q) \Rightarrow (\sim p \Rightarrow q)\Big) \equiv \Big(p \vee q\Big)$

$(p \Rightarrow \sim q) \Rightarrow (\sim p \Rightarrow q) \equiv \sim(p \Rightarrow \sim q) \vee (\sim p \Rightarrow q)$ table 7

$\equiv \sim(\sim p \vee \sim q) \vee (p \vee q)$ table 7

$\equiv (p \wedge q) \vee (p \vee q)$ D.M.

$\equiv ((p \wedge q) \vee p) \vee q$ Assoc. Prop.

$\equiv p \vee q$ table 6

(b) *(5 points)* $\Big(\Big(\sim p \wedge (p \vee q)\Big) \Rightarrow q\Big) \equiv \mathbb{T}$

$(\sim p \wedge (p \vee q)) \Rightarrow q$

$\equiv \sim(\sim p \wedge (p \vee q)) \vee q$ table 7

$\equiv (p \vee \sim(p \vee q)) \vee q$ D.M.

$\equiv (p \vee (\sim p \wedge \sim q)) \vee q$ D.M.

$\equiv ((\underbrace{p \vee \sim p}_{\mathbb{T}}) \wedge (p \vee \sim q)) \vee q$ Distr.

$\equiv (\mathbb{T} \wedge (p \vee \sim q)) \vee q$

$\equiv ((p \vee \sim q) \vee q)$ Id. Law

$\equiv p \vee (\sim q \vee q)$ Assoc

$\equiv p \vee \mathbb{T} \equiv \mathbb{T}$ Dom. Law

Problem 4. *(8 points) Satisfiability*
Determine whether the following compound proposition is satisfiable. If it is, provide at least one assignment of truth values which "satisfies" the compound proposition. Describe why the compound proposition is a tautology, contingency, or a contradiction.

$$\Big((\sim p) \vee (\sim q) \vee r\Big) \wedge \Big((\sim p) \vee q \vee (\sim s)\Big) \wedge \Big(p \vee (\sim q) \vee (\sim s)\Big)$$
$$\wedge\Big((\sim p) \vee (\sim r) \vee (\sim s)\Big) \wedge \Big(p \vee q \vee (\sim r)\Big) \wedge \Big(p \vee (\sim r) \vee (\sim s)\Big)$$

This is satisfiable if it can be T.

This is T iff

$\sim p \vee \sim q \vee r = T$ and $\sim p \vee q \vee \sim s = T$ and
$p = F$ ✓ $p = F$ ✓
$r = T$ ✓ $q = T$ ✓

$p \vee \sim q \vee \sim s = T$ and $\sim p \vee \sim r \vee \sim s = T$ and
$q = F$ ✓ $r = F$ ✓
$p = T$ ✓ $s = F$ ✓

$p \vee q \vee \sim r = T$ and $p \vee \sim r \vee \sim s = T$
$r = F$ ✓ $s = F$ ✓
$p = T$ ✓ $p = T$ ✓

None are restrictive, make choices and see what happens.

The assignment $p = F$, $q = F$, $r = F$, $s = F$ satisfies the compound proposition.